

LARGEMOUTH BASS AQUACULTURE

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First published 2019

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Published by
5M Publishing Ltd,
Benchmark House,
8 Smithy Wood Drive,
Sheffield, S35 1QN, UK
Tel: +44 (0) 1234 81 81 80
www.5mpublishing.com

A Catalogue record for this book is available from the British Library

ISBN 9781789180480

Book layout by Servis Filmsetting Ltd, Stockport, Cheshire
Printed and bound in Wales by Gomer Press Ltd
Photos by the authors unless otherwise indicated
Bass drawing on the front cover by Rick Hill, Kentucky Division of Fish and Wildlife Resources
Front cover design by Patrick Tidwell

CHAPTER 4

Environmental requirements for the culture of largemouth bass

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4.1 Introduction

Largemouth bass are adapted to temperate, freshwater environments. Since temperate aquatic environments characteristically exhibit extensive variability in physiochemical characteristics, it is not surprising that largemouth bass are tolerant to wide ranges of specific characteristics. In this chapter, the preferences and limits of largemouth bass to water quality characteristics important to aquaculture will be summarized.

4.2 Temperature

Temperature is the predominant physiochemical characteristic in aquaculture operations. It affects the amount of oxygen a given amount of water will hold, the rate of photosynthesis and decomposition, the ionization of ammonia, and the metabolic rate of the fish. Changing temperatures are also major cues for reproduction. Fry (1947, 1971) characterized the role of temperature as an environmental controller, and as such, should be given primary consideration when siting an aquaculture operation.

Largemouth bass eggs spawned at 17–21°C exhibited good hatching ($\geq 90\%$) when incubated between 13 and 24°C. By slowly (0.2°C/h) changing from spawning to incubation temperature, the range of good hatching was extended to 10–27°C (Kelley, 1968). When tested under constant-temperature conditions ranging from 15 to 36°C, fry grew best in 25–27°C water (Coutant and DeAngelis, 1983).

Largemouth bass fingerlings (mean 9.1 g) were cultured in tanks for 97 days on a commercial salmonid diet at either 20, 26, or 32°C (Tidwell et al., 2003). Survival averaged 94% and was not affected by water temperature. Weight gain in the 26 and 32°C treatments was similar and averaged 647% while weight gain in the 20°C treatment averaged 325%. Across the three treatments, feed conversion ratio ranged from 1.0 to 1.2 and was lowest in the 26°C treatment. Condition

Table 4.1 Temperatures of importance to the culture of largemouth bass *Micropterus salmoides*.

Temperature (°C)	Action	References
2.0	Temperature at which 48% of Florida subspecies died after 5 days exposure	Carmichael et al., 1988
10.0–27.0	Good ($\geq 90\%$) hatching success when acclimated to hatching temperature at 0.2°C/h	Kelley, 1968
25.0–27.0	Best temperature range for fry growth	Coutant and DeAngelis, 1983
26.0	Best temperature for fingerling growth based on a 97-day growth trial	Tidwell et al., 2003
28.1–28.6	Best temperature for growth based on temperature preference	Díaz et al., 2007
37.3	Chronic thermal maximum for northern subspecies	Fields et al., 1987
39.2	Chronic thermal maximum for Florida subspecies	Fields et al., 1987

factor (K) ranged from 1.3 to 1.4 and was lowest in the 20°C treatment. In aggregate, these results indicate that 26°C is the best temperature of the three tested in which to grow largemouth bass fingerlings. A study by Díaz et al. (2007) using a temperature preference approach estimated the optimum temperature for growth to be between 28.1 and 28.6°C.

Two subspecies of largemouth bass, the northern largemouth bass (*Micropterus salmoides salmoides*) and the Florida largemouth bass (*Micropterus salmoides floridanus*), are recognized and cultured in the USA (Philipp et al., 1985). As the names imply, one subspecies is native to more northern areas while the other is native to more southern areas. The two subspecies do differ from each other in characteristics that may be important to aquaculture. For instance, when water temperature is reduced to 2°C (at a rate of 1°C/day) and then held for 5 days, about one-half of the exposed Florida largemouth bass fingerlings died while only 4–5% of northern fish died (Carmichael et al., 1988). Interestingly, hybrids of the subspecies (both originals and reciprocals) exhibited low-temperature tolerance similar to the northern subspecies. When water temperature was increased at a rate of 1°C/day, the northern subspecies was the most susceptible to high temperature while the Florida subspecies was most resistant and the hybrids were intermediate (Fields et al., 1987). Table 4.1 summarizes temperatures of importance to the culture of largemouth bass.

4.3 Dissolved oxygen

Fish, like all animals, require oxygen to transfer energy efficiently from molecules making up food to molecules that can transfer the energy to the host of physiological processes within the organism. Most oxygen in culture water comes from two sources. First, oxygen diffuses from the atmosphere down a pressure gradient into the water (mechanical aerators simply enhance this process). Second, oxygen is produced within the water by photosynthetic activity of algae. Some intensive aquaculture systems may also add pure oxygen. Oxygen dissolved in the water will enter a fish by diffusing down a pressure gradient across the gills into the blood of the fish. Once in the fish, the oxygen is transported to the tissues where it is needed, at which point it further

diffuses down a pressure gradient into the tissue that will consume it. It should be noted that the driving force for delivering oxygen to the tissues is passive (diffusion down a pressure gradient). The pressure gradient is maintained by a high concentration of oxygen in the water (because of constant inputs) and a low concentration of oxygen in the tissues (because of constant consumption). There is no active uptake system for oxygen. Hence, acceptable environmental oxygen levels must be maintained at all times or the fish will not have adequate oxygen to function to its potential (Colt and Tomasso, 2001).

When discussing oxygen concentrations required for fishes, temperature must be an integral part of the discussion for two reasons. First, increasing temperature increases the metabolic rate of fishes, which is reflected in increased oxygen needs (Fry, 1947, 1971). Second, warmer water holds less oxygen than cooler water (Table 4.2 this volume, discussed in Colt and Tomasso, 2001). Taken together, fish in warmwater have a higher oxygen requirement in an environment with a decreased pressure gradient with which to deliver the oxygen to the tissues. For example, oxygen consumption of largemouth bass fingerlings acclimated for 30 days to 20, 23, 26, 29, and 32°C demonstrated a linear increase in oxygen consumption (from 46 to 86 mg O₂/kg/h) over the range of acclimation temperatures, an 87% increase (Díaz et al., 2007).

Largemouth bass embryo development and egg hatching as a function of temperature and dissolved oxygen concentrations were investigated in a comprehensive study by Dudley and Eipper (1975). Their results indicated that dissolved oxygen concentrations should be above 2.0, 2.1, and 2.8 mg/l for incubation temperatures of 15, 20 and 25°C, respectively, for successful hatching. However, the investigators observed many abnormally developed larvae. A further analysis of the results suggested that dissolved oxygen concentrations of 2.0, 2.5, and 3.5 mg/l for incubation temperatures of 15, 20, and 25°C, respectively, are adequate to reduce the incidence of abnormally developed larvae to the level found in incubation water containing 90% oxygen saturation. The Dudley and Eipper study also indicated that vertical movement of the hatching chamber in the water column was detrimental to hatching.

Critical oxygen levels (the lowest dissolved oxygen level at which fish consistently survived for 24 h) for largemouth bass fingerlings were determined by Moss and Scott (1961). Fish acclimated to 25, 30, and 35°C demonstrated critical oxygen levels of 0.92, 1.19, and 1.40 mg/l, respectively. These values fit well with residual oxygen concentrations (dissolved oxygen concentrations at time of death from hypoxia) of ~0.7 mg/l for largemouth bass fingerlings acclimated to 22°C (Carmichael et al., 1988).

Table 4.2 Dissolved oxygen saturation in pure water at selected temperatures under an air atmosphere at 760 mm Hg pressure (modified from Boyd, 1979).

Temperature (°C)	Dissolved oxygen (mg/l)	Temperature (°C)	Dissolved oxygen (mg/l)
8	11.5	22	8.5
10	10.9	24	8.3
12	10.4	26	8.0
14	10.0	28	7.8
16	9.6	30	7.5
18	9.2	32	7.3
20	8.8	34	7.1

Table 4.3 Dissolved oxygen concentrations of importance to the culture of largemouth bass *Micropterus salmoides*.

Dissolved oxygen (mg/l)	Action	Reference
2.0, 2.5, 3.5	Minimum oxygen levels in 15, 20, and 25°C water, respectively, for good hatching success and normally developed larvae	Dudley and Eipper, 1975
0.92, 1.19, 1.40	Critical levels of dissolved oxygen (the lowest level at which fish consistently survived for 24 h) at 25, 30, and 35°C	Moss and Scott, 1961
0.7	Residual oxygen concentration (oxygen concentration at time of death) at 22°C	Carmichael et al., 1988
1.5	Residual oxygen concentration at 22°C in the presence of 0.7 mg/l un-ionized ammonia	Carmichael et al., 1988
1.6	Reduced growth and feed conversion at 26°C	Stewart et al., 1967
4.0	Concentration at which fingerling growth and feed conversion did not differ from fish in higher oxygen concentrations at 26°C	Stewart et al., 1967

In a short-term (11–15 day) laboratory study at 26°C (Stewart et al., 1967), largemouth bass fingerling growth was shown to increase with increasing dissolved oxygen levels in a near-linear fashion up to air saturation levels (~8.2 mg/l dissolved oxygen). Feed conversion also improved as dissolved oxygen levels increased. Dissolved oxygen levels as low as 1.6 mg/l did not affect survival but did affect growth and feed conversion. The authors indicated 4 mg/l as an oxygen level above which growth and feed conversions did not differ markedly. Interestingly, dissolved oxygen levels above air saturation levels (achieved using pure oxygen) had a detrimental effect on growth and feed conversion. The high pressure gradient in the supersaturated water may have suppressed ventilation (breathing) activity of the fish because oxygen was flowing quickly into the blood. However, a consequence of reduced ventilation is reduced release of carbon dioxide which can lead to metabolic pH disturbances in the fish (Wood, 1991) and perhaps have an impact on growth and feed conversion.

Table 4.3 summarizes dissolved oxygen levels of importance to largemouth bass. It appears that largemouth bass are fairly tolerant of low dissolved oxygen concentrations. However, oxygen levels affect fish health and immune systems as well as survival, growth and feed conversion. It is advised that culturists design production systems to maintain dissolved oxygen levels well above minimums discussed here.

4.4 Dissolved solids

Surface and ground waters contain dissolved substances that are derived from the environment as the water passes through, on and below the surface of the Earth. When speaking of freshwater, the term “total dissolved solids” is used to denote all dissolved solids, regardless of the specific nature, found in a sample. When speaking of sea water, the term “salinity” is used to describe the total concentration of a very specific set of ions and complex ions that exist in specific ratios to each other (Spotte, 1979). The terms salinity and total dissolved solids are often used interchangeably, but doing so can be misleading.

Table 4.4 Major ions, their chemical forms, and concentrations found in full-strength sea water (adapted from Bowen, 1966, and Spotte, 1979).

Ion	Major form	mg/l	Ion	Major form	mg/l
Chloride	Cl ⁻	19,000	Potassium	K ⁺	380
Sodium	Na ⁺	10,500	Bromide	Br ⁻	65
Magnesium	Mg ⁺⁺	1,350	Carbon	CO ₃ H ⁻	28
Sulfur	SO ₄ ⁻⁻	885	Strontium	Sr ⁺⁺	8
Calcium	Ca ⁺⁺	400	Silicon	Si(OH) ₄	3

Within freshwater, two important ions are calcium and magnesium. Collectively, they make up the hardness of the water. Hardness is expressed as milligram per liter of equivalent calcium carbonate (Boyd, 1979). The environmental calcium is especially important in fish culture because it helps reduce water and ion flux across the gills, reducing the energy requirements to maintain homeostasis in the fish (Weirich et al., 1992; Grizzle et al., 1993). Water hardness requirements for the culture of freshwater fishes are not well defined. The recommended hardness for the culture of striped bass *Morone saxatilis* juveniles is 150 mg/l as calcium carbonate (Hall, 1991). Largemouth bass appear to do well in a range of hardnesses.

Seawater contains 35 g/l sea salt (Spotte, 1979). Sea salt contains over 60 elements. Table 4.4 lists some of the elements and their major forms found in seawater. In the tidal areas of rivers entering estuaries, it is common to find an extended continuum of sea water to freshwater. Largemouth bass are often found in the continuum (Meador and Kelso, 1989, 1990; DeVries et al., 2015). Freshwater fishes do have a limit to the salinity they can tolerate for extended periods of time. This limit corresponds to 10–12 g/l salinity, the point where water begins leaving the fish osmotically across the gills. Long-term survival above 10–12 g/l requires physiological mechanisms characteristic of estuarine and seawater-adapted fishes (Colt and Tomasso, 2001). Salinity is expressed as parts per thousand, grams per liter, or per mille (‰). For practical purposes, all mean the same thing (gram of sea salt per liter of solution).

In a laboratory study by Meador and Kelso (1990), largemouth bass (185–289 mm total length) were collected from a freshwater lake and a 2 g/l salinity marsh. The fish were then cultured for 120 days at 0, 4, 8, or 12 g/l salinity (22°C). Fish from the freshwater lake grew equally well in 0 and 8 g/l salinity. Growth was lower in 8 g/l, and no fish survived the study in 12 g/l. Fish from the marsh grew equally well in 0, 4, and 8 g/l salinity, while no fish survived the study in 12 g/l. These results indicate the potential to grow largemouth bass in low-salinity water.

4.5 pH, carbon dioxide, and alkalinity

The negative logarithm of the hydrogen ion concentration of a solution is defined as pH. It presents on a scale of 1 to 14. Solutions with a pH below 7 are acidic, solutions with a pH above 7 are basic (alkaline), and solutions with a pH of 7 are neutral. The further the number is from seven, the stronger the acid or base. The sensitivity of largemouth bass to pH is not well defined. However, anecdotal evidence indicates that they do well in the range of 6.5 to 9.0 recommended for hatchery operations (Piper et al., 1982).

When discussing pH and culture conditions, it is important to distinguish between chronic and transient exposure. During pond culture, fish are typically exposed to a daily cycle of pH with lowest pH just after sunrise and highest pH just before sunset (see below in the discussion of alkalinity). In raceway culture, using well or stream water, or recirculating culture, the fish may be exposed to constant pH values. Largemouth bass appear to do well in pond culture and can tolerate a wide range of pH values, including short term exposures of up to 10 (characteristic of low alkalinity ponds). However, continuous exposure to high or low pH water may have detrimental effects.

Two major detrimental effects can be attributed to extreme culture pH values. At the low end, bicarbonate (H_2CO_3) in solution will dehydrate to form free carbon dioxide (CO_2). This process increases in rate as pH falls below 7.0 (Boyd, 1979). As the partial pressure of carbon dioxide increases in the culture water, it inhibits the excretion of carbon dioxide from the fish by way of the gills. As carbon dioxide builds up in the blood, it interferes with oxygen transport. At the high end, pH will increase the fraction of ammonia in its toxic form (see below discussion on ammonia). Consequently, extreme pH situations, especially during continuous exposure, need to be monitored and managed. Little work has been conducted to understand the sensitivity of largemouth bass to carbon dioxide. However, one study using corticosteroid hormones as indicators of stress reported that largemouth bass appeared unstressed (pH ~7.3) when exposed to 35 mg/l carbon dioxide but slightly stressed when exposed to 135 mg/l (Carmichael et al., 1984).

An example of a management strategy is one used in the author's lab for recirculating systems. In this case, we set pH limits of 7.5 and 8.0. The range is applied to both freshwater and seawater systems and considers three aspects of pH and culture – low pH increases free carbon dioxide concentrations, high pH increases the fraction of ammonia in its toxic form, and the pH range that supports efficient biofilter function (Spotte, 1979). We monitor pH daily, or every few days. Functioning biofilters will suppress pH as hydrogen ions are released in the nitrification process. Buffering capacity is less in freshwater than saltwater, so pH will fall faster in freshwater systems. When pH reaches 7.5, we add adequate sodium bicarbonate to raise the pH to near 8.0.

The total titratable base in an aqueous solution is referred to as total alkalinity (Boyd, 1979). It is expressed as equivalent calcium carbonate. In ponds, alkalinity interacts with algal photosynthesis in that periods of high net carbon dioxide production by algae (evening) tend to reduce pond pH while periods of high net algal carbon dioxide consumption (daytime with sunshine) tend to allow pH to rise. The degree of the daily pH shift is partly a function of the alkalinity – high alkalinity ponds have a higher buffering capacity and lower pH shift, lower alkalinity ponds have a lower buffering capacity and broader pH shift (Boyd and Tucker, 2014). The alkalinity of culture water has no direct impact on the culture animals. It is the pH that is important, so managing alkalinity is an indirect way of managing pH. Managing alkalinity by adding lime to ponds is discussed in Boyd (1979) and Boyd and Tucker (2014).

4.6 Nitrogenous wastes

Fish feed contains protein to support fish development, growth and health. When feed is introduced into a pond or other culture unit, most (hopefully) of the feed is consumed by the fish. The remainder is consumed by the decomposer community in the pond. By either route, part of

the protein is retained in additional biomass (fish or decomposer growth) and part is used as an energy source. Largemouth bass, being carnivores, will use more protein for energy than omnivores, such as catfish. Hence, catfish feeds have a lower level of protein than feed for carnivorous fishes. When protein is metabolized for energy, it is first reduced to its component amino acids which, in turn, are deaminated. The deamination step produces ammonia, which is very toxic to animals and must be excreted. Once in the culture water, ammonia is normally removed by the processes of the nitrogen cycle in ponds, which includes the process of nitrification, or removed simply by nitrification in recirculating systems (Spotte, 1979; Boyd and Tucker, 2014). If the nitrogen cycle and nitrification are working correctly, the concentration of ammonia in the culture water remains low. However, if feed inputs exceed the capacity of the ammonia removal systems or if some other factors affect the systems (for example, rapid temperature change), a buildup of ammonia may occur in the culture system.

In aqueous solution, ammonia exists in two forms – ammonia (NH_3) and ammonium (NH_4^+). Together, these two chemical species are referred to as total ammonia. NH_3 is the toxic form of ammonia because it is capable of diffusing across the gill membrane (Tomasso, 1994; Wright and Wood, 2009). The toxicity of ammonia to fishes depends largely on the pH of the culture water because the pH controls the proportion of the total ammonia that is in the toxic form (Table 4.5). Temperature has a lesser effect on ionization. As can be seen in Table 4.5, the percent of ammonia in the toxic form rises sharply as pH rises above 8, so ammonia levels should be carefully monitored when culturing fish at constant pH values above 8. Transient pH values above 8 such as during daily pH cycling in ponds may not be detrimental because the animals have an opportunity to excrete ammonia during the periods of lower pH.

Limited studies (unpublished, Carmichael and Tomasso) indicate that largemouth bass have a 72 h median-lethal concentration (the concentration of a poison that will kill one-half of exposed animals in 72 h) of 15.3 mg/l total ammonia-nitrogen at pH 8.0 and temperature 25°C. Of this total, 0.82 mg/l is in the un-ionized, or toxic, form (see Table 4.5). Recommendations for “safe” levels vary, but a rule of thumb is to keep ambient levels no higher than one-tenth and preferably below one-twentieth of lethal levels (0.082 and 0.041 mg/l, respectively).

The process of nitrification is mediated by nitrifying bacteria (Spotte, 1979; Boyd and Tucker, 2014). During this process, ammonia is oxidized to nitrite (NO_2^-) and then to nitrate (NO_3^-). In a balanced pond or recirculating system, ammonia and nitrite concentrations are low and nitrate either accumulates or is removed by assimilation into aquatic plants. As noted above, perturbations of the nitrification process may lead to increases in ammonia concentrations. It

Table 4.5 The percent of total ammonia in aqueous solution that is in the un-ionized (NH_3) form at five pH values and three temperatures (modified from Emerson et al., 1975)

pH	Percent in un-ionized form				
	10°C	15°C	20°C	25°C	30°C
6	00.019	00.027	00.040	00.057	00.081
7	00.186	00.273	00.396	00.566	00.799
8	01.830	02.670	03.820	05.380	07.460
9	15.700	21.500	28.400	36.300	44.600
10	65.100	73.300	79.900	85.100	89.000

may also lead to increases in nitrite concentrations. Nitrite, like ammonia, may be toxic to fish. Conversely, nitrate is not very toxic.

Nitrite is highly toxic to many species of fishes (Tomasso, 1994). However, largemouth bass is not one of them. Palachek and Tomasso (1984) reported a 96 h median-lethal concentration to largemouth bass fingerlings of 140 mg/l nitrite-N, compared to 7.1 mg/l for channel catfish (*Ictalurus punctatus*). The basis for the large interspecies difference in toxicity was later shown to be related to differences in chloride-uptake mechanisms (Tomasso and Grosell, 2005). It is doubtful that nitrite will reach detrimental levels in largemouth bass culture systems. The most obvious clinical sign of nitrite toxicity is the presence of brown blood brought about by nitrite-induced oxidation of the iron in hemoglobin (Tomasso, 1994). If a culturist suspects nitrite toxicity, a brownish coloration to the blood or gills would serve as a presumptive diagnosis.

No work has focused on nitrate toxicity to largemouth bass. However, work on the closely related Guadalupe bass (*Micropterus treculi*) showed a 96 h median-lethal concentration of 1,261 mg/l nitrate-nitrogen (Tomasso and Carmichael, 1986). This value fits well with other toxicity values for freshwater fishes reviewed in Camargo et al. (2005). Nitrate does not appear to be a consideration from a toxicity perspective; however, nitrate is a fundamental plant nutrient, so high environmental nitrate levels can be expected to affect phytoplankton growth in any culture system receiving sunlight.

4.7 Gas supersaturation

Atmospheric gases over water tend toward an equilibrium with gases dissolved in the water. Biological and chemical activity in the water consume dissolved gases (particularly oxygen), usually keeping the total gas pressure in the water below that of the air above it and providing a pressure gradient for further diffusion from the atmosphere. Occasionally, culture water may become gas supersaturated – the total gas pressure in the water is higher than in the atmosphere above it. Gas supersaturation is an unstable condition in which the excess gas in the water is prone to form bubbles (much like the bubbles that form when a carbonated drink is opened). If these dissolved gases diffuse across the gills into the fish, and then bubbles form, capillary and small vessel circulation will be inhibited (termed gas bubble disease) resulting in death of the culture animals.

Gas bubble disease is not usually a problem in ponds given the large surface area that is the interface between the water and atmosphere. Also, mechanical and wind aeration promote degassing from water to atmosphere down the pressure gradient. Problems with supersaturation tend to occur in hatcheries, holding facilities and recirculating systems where water is pumped or heated. Heating reduces the ability of water to hold dissolved gases, so increasing the temperature of saturated water puts it in a supersaturated condition (this is why bubbles form on the sides of a saucepan when heating water on a stove). Pumps with air leaks on the low-pressure side will draw in air and then force it into the water on the high-pressure side resulting in supersaturation. Another common way water is supersaturated in fish production facilities is filling empty pipes with water with the end valves closed, pressurizing the air in the pipes. Colt and Tomasso (2001) and Hargreaves and Tomasso (2004) discuss supersaturation in more detail.

No controlled studies of the effects of gas supersaturation on largemouth bass were found. However, anecdotal reports from hatcheries clearly indicate that largemouth bass are susceptible to gas supersaturation and exposure should be avoided.

Largemouth bass are also susceptible to swim bladder stress syndrome (Carmichael and Tomasso, 1984), a malady that may be mistaken for gas bubble disease. This syndrome is characterized by overinflated swim bladders that may cause the fish to broach. It is distinct from gas bubble disease in that no emboli form in the blood. The etiology of swim bladder stress syndrome is not clear; however, it appears to be more related to handling stress than gas supersaturation.

A final consideration when discussing gas supersaturation is oxygen supersaturation, which can occur when pure oxygen is being used to maintain proper oxygen levels in water. This often occurs in transport tanks when oxygen is being metered into the tanks at too high a rate. Measured oxygen levels can reach two or more times the saturation level for aeration systems; however, the total gas pressure does not exceed atmospheric pressure though because oxygen is displacing nitrogen in the water. Oxygen supersaturation can cause developmental problems if transporting eggs, embryos or fry; however, it is generally not a problem when transporting fingerlings.

4.8 Hydrogen sulfide

Hydrogen sulfide (H_2S) is a toxic gas produced by way of anaerobic activity of bacteria. It is sometimes found in well water and in anaerobic pond bottoms where organic material has collected. A common problem in warmwater pond culture is the release of hydrogen sulfide from the sediments when fingerling ponds are drained to collect fish. Raking pond bottoms and allowing the ponds to remain empty between production cycles reduces the accumulation of organic material and incidence of hydrogen sulfide release.

Maintaining aerobic conditions will promote the oxidation of sulfides to sulfates, which are not toxic. Hydrogen sulfide exists in a pH- and temperature-dependent equilibrium with the HS^- ion. The non-ionic species is the toxic form because it is gill permeable (similar to ammonia). At pH 7 and temperature 25°C, about one-half of the sulfide is in the toxic form (Table 4.6). At a pH of 9, only about 1% is (Wedemeyer, 1996). The USEPA (1986) recommends a maximum

Table 4.6 Effect of pH on the percentage of total hydrogen sulfide and total chlorine in the toxic forms (H_2S and $HOCl$, respectively).

pH	Percent in toxic forms	
	H_2S	$HOCl$
5	99	100
6	91	97
7	51	76
8	9	23
9	1	3

Source: Hydrogen sulfide information adapted from Wedemeyer (1996) at 25°C. Chlorine information adapted from Boyd and Tucker (2014) at an unreported temperature.

safe exposure level for aquatic animals of 0.002 mg/l $\text{H}_2\text{S-S}$. Tucker (1993) suggests half the USEPA maximum.

4.9 Chlorine

Chlorine is a very effective disinfectant. However, it is also very toxic to aquatic animals. Chlorine can be applied as a liquid, gas, or solid. Regardless of the form of application, an equilibrium forms in the water between the highly toxic hypochlorous acid (HOCl) and less toxic OCl^- . The pH of the water determines the nature of the equilibrium with more HOCl present at high pH values (Table 4.6). The equilibrium is at molar equality at a pH of about 7.5 (Boyd and Tucker, 2014).

Recommended maximum exposure levels for aquatic animals range from 0.001 to 0.10 mg/l total chlorine (summarized in Colt and Tomasso, 2001). Chlorine can be removed by activated carbon, aeration, or sunlight. Simple-to-use test kits are commercially available and should be used after any process in which chlorine is used.

4.10 Copper sulfate

Copper sulfate is commonly added to aquaculture ponds for the control of algae and external parasites. Caution must be exercised when using copper sulfate because copper is very toxic to fish and water chemistry has a large impact on both toxicity and effective concentrations. For example, the 96 h median-lethal concentration of copper (from copper sulfate) to channel catfish increases 20-fold (0.05 to 0.98 mg/l) as the alkalinity is increased from 16 to 240 mg/l as CaCO_3 (Straus and Tucker, 1993). The Biotic Ligand Model (DiToro et al., 2000) was developed to integrate the effects of multiple water quality characteristics on both toxicity and effectiveness. Calculating effective and safe levels of copper sulfate for pond application is complex. Inexperienced culturists are encouraged to contact their local extension office for guidance.

4.11 Formalin

Formalin is a water solution containing formaldehyde at saturation level (about 37%). Since malachite green was banned from use, formalin has become the standard antifungal treatment and preventative for fish eggs. It is also used for the treatment of ectoparasites on fish. The 24 h median-lethal concentration for formalin to largemouth bass fingerlings is about 140 $\mu\text{l/l}$. For ectoparasite control, fish are usually exposed to high concentrations (150–250 $\mu\text{l/l}$) for about 1 h.

An interesting aspect of formalin use with largemouth bass is that fingerlings are about twice as resistant to formalin as are tadpoles (*Rana berlandieri*). A common problem during largemouth bass fingerling production is that large numbers of tadpoles are harvested from ponds along with the fish – at times the biomass of tadpoles exceeds the biomass of fingerlings. Separating the fish and tadpoles is time-consuming and stressful to the fish. Carmichael and Tomasso (1983) demonstrated that separation could be accomplished by exposing mixed groups of largemouth bass

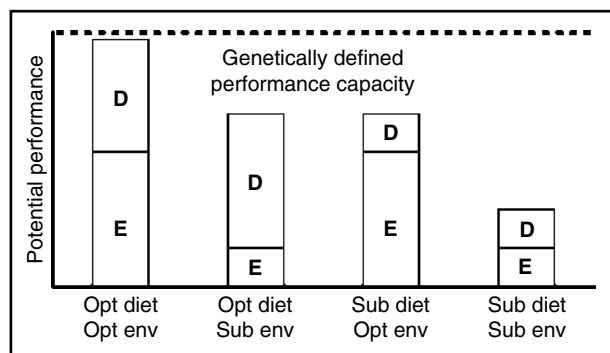


Figure 4.1 Conceptualization of the relationship among genetically defined performance capacity (dashed line), diet, and environment. Suboptimal diets (sub diet) and suboptimal environments (sub env) can reduce performance compared to optimal diets (opt diet) and optimal environments (opt env). The relative contributions of diet and environment shown here are for illustration and not based on experimentation.

fingerlings and tadpoles to 200 µl/l formalin for one hour followed by flushing with 6.9 turnovers per hour. By the end of 24 h, all of the tadpoles were dead and washed toward the drain end of the tank. As with copper sulfate, inexperienced culturists are advised to contact their local extension office for guidance and the latest regulations regarding the use of formalin on food-fish.

4.12 Conclusion

The largemouth bass is a hardy temperate species that is tolerant of a range of water quality. This chapter has highlighted several aspects of the environmental requirements of the species. Several other resources (Williamson et al., 1993; Colt and Tomasso, 2001; Hargreaves and Tomasso, 2004; Boyd and Tucker, 2014) provide additional details and insight.

This chapter has focused on the environment for producing largemouth bass. However, it is important to put environment into perspective. Schreck (1981) used the concept of performance capacity to describe the relationship of a fish to its environment. Essentially, an animal has a genetically defined maximum capacity to perform (survive, grow, reproduce, and so on). Maximum performance capacity is reduced by suboptimal environments or stress. This concept was later put into an aquaculture context (Tomasso, 1995). Figure 4.1 shows the performance capacity concept and how two broad categories (environment and nutrition) can have an impact on performance. In summary, maintaining environments (and diets) as close to optimum as possible will result in better performance (survival, growth, reproduction, feed conversion).

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